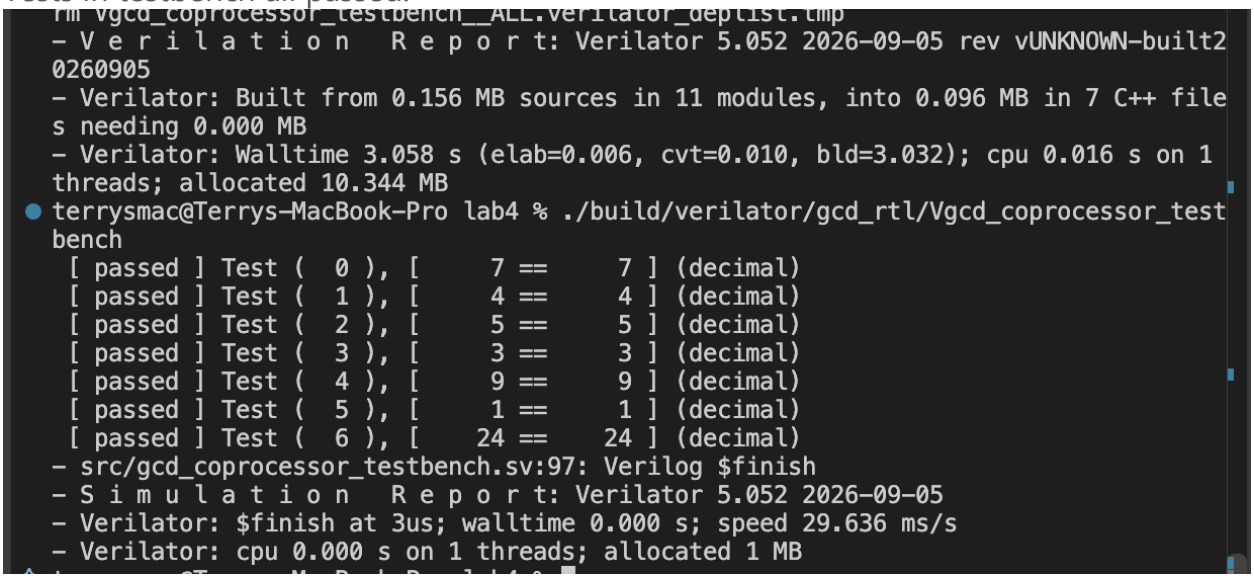
Question 1: GCD Coprocessor Design

a. Submit your code (`gcd_coprocessor.sv` and `fifo.sv`) and show that your code works (VCS output screenshot is fine)
Tests in testbench all passed:



Question 2: Interpreting PAR Timing Reports

Note: If you are using VSCode, a helpful extension to view the timing reports is `Gzip Decompressor`. The timing reports for the first part can be found at:

```
build/par-rundir/timingReports/gcd_coprocessor_preCTS_all.tarpt.gz
build/par-rundir/timingReports/gcd_coprocessor_postCTS_all.tarpt.gz
build/syn-rundir/reports/final_time_ss_100C_1v60.setup_view.rpt
```

a. What is the critical path of your design pre-CTS and post-CTS? Is it the same as the post-synthesis critical path?
critical path of pre-CTS is from flipflop_1030 to flipflop_1060
critical path of post-CTS is from flipflop_1030 to flipflop_1046
critical path in pre-CTS and post-CTS are not the same

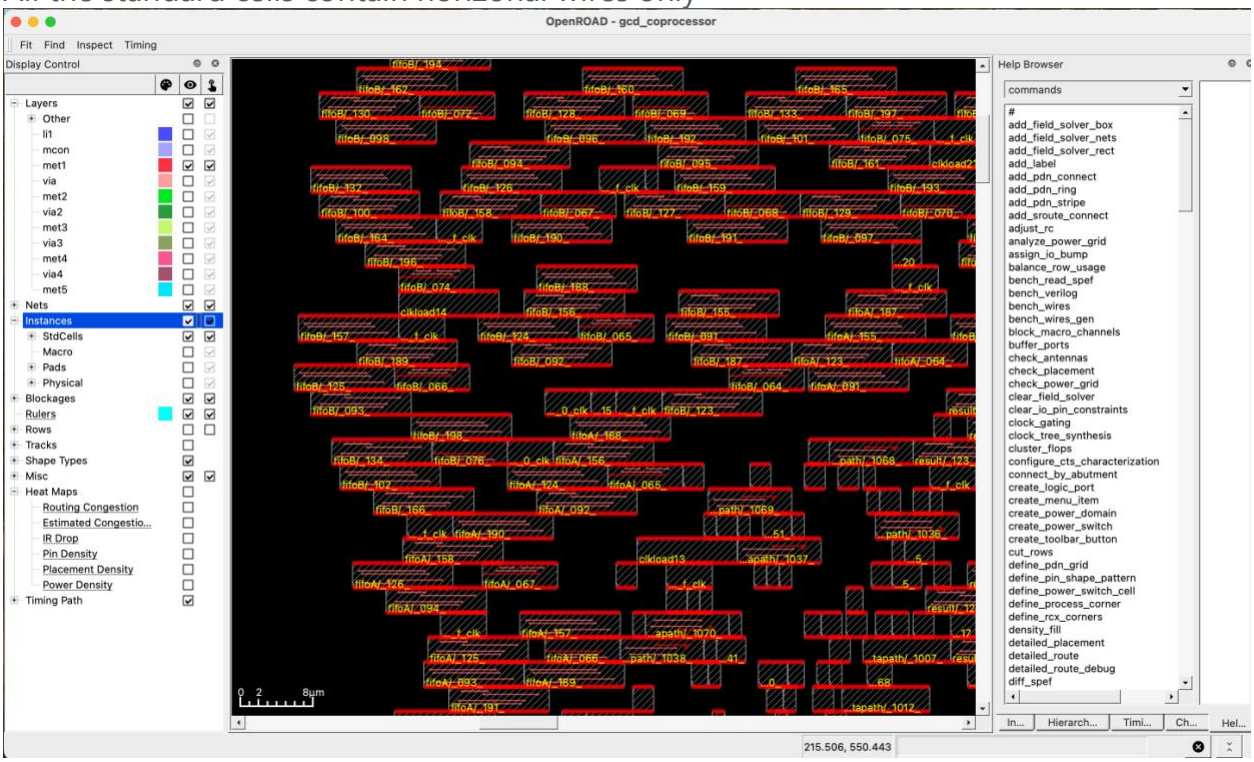
b. Look in the post-CTS timing report. Find a path inside which the same type of cell is used more than once. Identify the delay of those instances of that common cell. Can you explain why they are different?
In the post-CTS timing repot, the standard cell sky130_fd_sc_hd__o311ai_4

appears multiple times. The delay in this cell vary from 0.13ns to 0.29ns. The difference in delay is because propagation delay depends on electrical conditions, including input slew, output load capacitance

c. What is the skew between the clock that arrives at the flip-flops at the beginning and end of the post-CTS critical path? Does this skew help or hurt the timing margin calculation? As a refresher, please read this article about clock skew.
The clock skew is 0.01ns.
In post-CTS timing report, the launch clock network delay is 0.55ns, while capture clock arrival is at 0.54 ns. There is a 0.01 ns delay in between.

Question 3: Understanding PAR Steps

a. Visualize standard cell routing by enabling "Cell" in the right panel of Innovus. Visualize only the M1 layer and submit a screenshot of the entire layout (you can press `f` to zoom to fit). Next, zoom in a bit so you can see the individual traces. What pattern do you see with the M1 traces?
All the standard cells contain horizonal wires only



b. Now, reenable all the layers. You should see yellow triangles at the bottom of your design which are the IO pins to your module. Find an input pin (triangle

facing upwards) and describe what part of the input it carries (hint: click on the trace it is attached to). Submit a screenshot of this and the name of the input wire.

operands_bits_A[0]:

